



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Course Title	Enterprise Architecture				
Course Code:	SYS1003	Credit Value:	3	Credit Hours:	42
Programs:	BAGC I.T. Business Analysis BTHP I.T. Business Analysis - Hanson				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will explore the design, selection, implementation, and management of enterprise I.T. solutions. This course primarily focuses on applications, infrastructure, and their fit within businesses. Students learn frameworks and strategies for infrastructure management, system administration, data/information architecture, and examine business from an I.T. architecture point of view. These topics are addressed both within the organization and extending beyond the organization, with attention paid to managing risk and security. Students also improve their ability to communicate technology architecture strategies concisely to a general business audience. Additionally, students will study how the role of Enterprise Architecture (E.A.) is to architect and supervise this transformational journey.

Date: June 19, 2020

Prepared By: Nathan Abourbih

Approved by: 
BRIAN LOBBAN
Dean, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2020, Winter 2021, Spring 2021

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
I.T. Business Analysis	College Standards 1. Plan and manage the requirements gathering activities throughout the requirements process for changes to a business system 2. Analyze the information collected during the requirements elicitation to identify gaps and define the capabilities of an acceptable solution using appropriate methods, tools, techniques and documentation.
I.T. Business Analysis - Hanson	Ministry Standards 1. Plan and manage the requirements gathering activities throughout the requirements process for changes to a business system. 2. Analyze the information collected during the requirements elicitation to identify gaps and define the capabilities of an acceptable solution using appropriate methods, tools, techniques and documentation.

COURSE CURRICULUM**Topics/Concepts Covered in This Course**

- Goals and Benefits of Enterprise Architecture (EA)
- Core Activities of Enterprise Architects
- EA Framework
- EA Maturity Models
- Foundations of Collaborative EA
- Lean and Agile EA
- Building Blocks of EA 2.0
- Taking Collaborative EA Forward

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Rationalize the need for collaborative Enterprise Architecture (EA).	1.1 List goals and benefits of Enterprise Architecture (EA). 1.2 Define EA through four dimensions, perspective, governance, strategy, transformation. 1.3 List common challenges and failures for the caricatures/actor roles in each of the four dimensions, perspective, governance, strategy, transformation.

Learning Outcome	Objectives
	1.4 Analyze the effectiveness of Enterprise Architecture through four dimensions: perspective, governance, strategy, transformation. 1.5 Describe how a well-functioning EA within a business scenario, is able to prevent common problems such as pet programs, poor program management, over-ambition, technology risks. 1.6 Describe the challenges involved in balancing chaos and order as it relates to EA. 1.7 Describe the three guidelines of Collaborative EA. (1) Establish a lean set of processes and rules. (2) Adopt evolutionary problem solving. (3) Foster and moderate open participation.
2. Describe what is Enterprise Architecture.	2.1 List several definitions that attempt to encapsulate a consensus definition of EA. 2.2 Describe the meaning of architecture and list its key characteristics (modeling, visualization, communication). 2.3 Apply architecture meaning in the context of Business-IT Management in an enterprise. 2.3.1 Define Enterprise Architecture (EA) 2.3.2 Define Enterprise Architecture Management (EAM) 2.4 List the applications and use cases of EA.
3. Provide the core activities of EA.	3.1 Define the IT strategy (EA-1). 3.2 Model the architectures (EA-2). 3.3 Evolve the IT Landscape (EA-3). 3.3.1 Rationalize applications. 3.3.2 Discuss general IT transformations. 3.3.3 Discuss Service-Oriented Architecture (SOA). 3.4 Assess and build capabilities (EA-4). 3.4.1 Describe competence development for Enterprise Architects. 3.4.2 Discuss the formalization process of Enterprise Architecture (who in the organization does what.) 3.4.3 Position EA teams within an organization structure. 3.5 Develop and enforce standards and guidelines (EA-5). 3.5.1 Classify technology usage in an enterprise as core, declining, emerging, or poor dogs. 3.5.2 Enforce standards and guidelines. 3.6 Discuss key processes of monitoring a project portfolio (EA-6). 3.6.1 Build an IT-focused project portfolio. 3.6.2 Audit a portfolio. 3.7 Lead or coach projects (EA-7). 3.8 Manage risk involved in IT (EA-8).
4. Contrast attributes of various enterprise architecture frameworks.	4.1 Define and describe what is an EA framework.

Learning Outcome	Objectives
	4.2 Describe the Zachman's framework for EA. 4.3 Discuss The Open Group Architecture Framework (TOGAF) EA. 4.4 Describe the Gartner Methodology (formerly META framework). 4.5 Discuss some roles and uses of EA frameworks.
5. Apply maturity models to EA.	5.1 Describe how assessments can be leveraged in the effective management of EA. 5.2 Develop and use a maturity model to formally assess enterprise deficiencies in EA. 5.3 Compare and research maturity framework adoption cases from industry: 5.3.1 Case of OMB (US Office of Management and Budget) EA assessment framework. 5.3.2 Architecture Capability Maturity Model of the US Department of Commerce. 5.3.3 EA Maturity Model by MIT Center for Information System Research 5.4 Discuss some of the limitations of maturity assessment (subjective, academic, manipulative, bureaucratic, superfluous, misleading) and describe how these can be overcome.
6. Perform the tasks related to collaborative EA	6.1 Develop a data gathering strategy; identify and analyze data to develop SMART requirements

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- not applicable

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- show respect for the diverse opinions, values, belief systems, and contributions of others.
- interact with others in groups in ways that contribute to effective working relationships.

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- Group Work
- Research
- Presentations
- Case Studies

Resources Required

Books

Stefan Bente, Uwe Bombosch, Shailendra Langade, *Collaborative Enterprise Architecture Enriching EA with Lean, Agile, and Enterprise 2.0 Practices*, 1st, Morgan Kaufmann, Elsevier
ISBN: 978-0-12-415934-1

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	40%
Applied Activities(G)	40%
Quizzes/Test/Exam	20%

ADDITIONAL INFORMATION

College

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Prior Learning Assessment and Recognition (PLAR)

Students may contact the Pathways office at pathways@cambriancollege.ca

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling or academic tutoring. Our team will also be able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode for cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://cambriancollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality Regulations**Lab Attendance**

Lab attendance is mandatory since lab grades include an evaluation of lab reports and lab performance. Lab performance can only be judged in person, requiring that students attend all scheduled labs.